

SARAH ARCOS

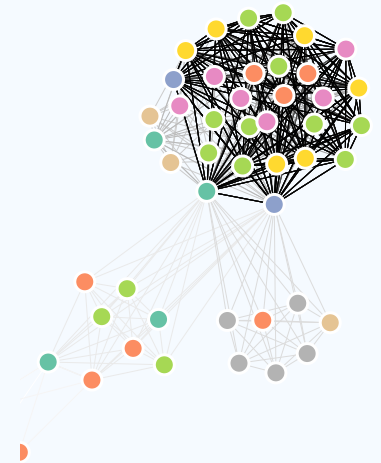
As a software developer I use my background in data science to build tools to help people explore, understand, and work with their data better. I have made visualizations viewed by hundreds of thousands of people¹, sped up query times for 25 terabytes of data by an average of 4,800 times², and built packages for R³ that let you do magic⁴.

EDUCATION

- 2020 • **PhD., Biostatistics**
Vanderbilt University Nashville, TN
- Dissertation: Network analysis and visualization for electronic health records data.⁵
 - Specialized in creating high-performance interactive visualization platforms
 - Developed algorithms for efficient real-time network data processing
- 2015 • **B.S., Mathematics, Statistics (minor C.S.)**
University of Vermont Burlington, VT
- Thesis: An agent based model of Diel Vertical Migration patterns of Mysis diluviana
 - Focused on computational efficiency, simulation optimization, and interactive model exploration

RESEARCH EXPERIENCE

- Current | 2015 • **Graduate Research Assistant**
TBILab (Yaomin Xu's Lab) Vanderbilt University
- Primarily working with large EHR and Biobank datasets.
 - Developing network-based methods to investigate and visualize clinically relevant patterns in data.
- 2018 | 2017 • **Data Science Researcher**
Data Science Lab Johns Hopkins University
- Building R Shiny applications in the contexts of wearables and statistics education.
 - Work primarily done in R Shiny and Javascript (node and d3js).
- 2015 | 2013 • **Undergraduate Researcher**
Rubenstein Ecosystems Science Laboratory University of Vermont
- Analyzed and visualized data for CATOS fish tracking project.
 - Head of data mining project to establish temporal trends in population densities of Mysis diluviana (Mysis).
 - Ran project to mathematically model the migration patterns of Mysis (honors thesis project.)

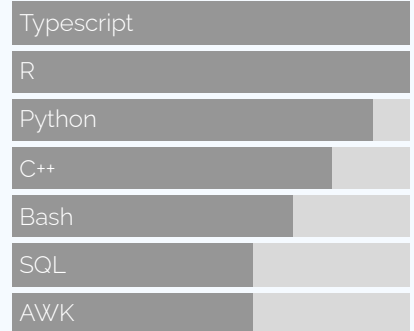


View this CV online with links at
[_https://robinvernooj.github.io/cv/_](https://robinvernooj.github.io/cv/)

CONTACT

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🌐 linkedin.com/in/nickstrayer

LANGUAGE SKILLS



Made with the R package [pagedown](#).

The source code is available on
github.com/nstrayer/cv.

Last updated on 2026-09-09.

- 2015

●

Human Computer Interaction Researcher
 LabInTheWild (Reineke Lab)

📍 University of Michigan

- Led development and implementation of interactive data visualizations to help users compare themselves to other demographics.
- 2014
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2013

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Undergraduate Researcher
 Bentil Laboratory

📍 University of Vermont

- Developed mathematical model to predict the transport of sulfur through the environment with applications in waste cleanup.
- 2013
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2012

●

Research Assistant
 Adair Laboratory

📍 University of Vermont

- Independently analyzed and constructed statistical models for large data sets pertaining to carbon decomposition rates.



INDUSTRY EXPERIENCE

- Current
|
2024

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Principal Software Engineer
 Posit

📍 Remote

- Architect and develop full-stack solutions for the Positron data science IDE
 - Worked across the Typescript, Python, and Rust codebase to build user-centric interfaces that balance performance with intuitive design
 - Collaborate across teams to ensure reliable, maintainable codebase architecture
 - Mentored junior developers on frontend best practices and code quality standards
- 2024
|
2023

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Senior Software Engineer
 Posit

📍 Remote

- Created and led development of ShinyUiEditor, a React-based drag-and-drop interface builder
 - Designed architecture for real-time previewing and component manipulation using custom psuedo-ast format that allowed translation into either R or Python from the same ast.
 - Spearheaded work to simplift and unify the UI layer of R and Shiny using custom webcomponents.
- 2023
|
2020

●

Software Engineer
 Posit

📍 Remote

- Part of team who created Shiny for Python, a ground-up rewrite of R's Shiny framework in Python

While most recently I have had the job title of "software engineer", I have worked in a variety of roles ranging from journalist to data scientist. Ultimately categorization is hard.

- 2016

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Data Journalist - Graphics Department
 New York Times

📍 New York, New York

- Reporter with the graphics desk covering topics in science, politics, and sport.
 - Work primarily done in R, Javascript, and Adobe Illustrator.
 - Developed interactive, data-dense visualizations viewed by hundreds of thousands of users

2015

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Engineering Intern - User Experience
 Dealer.com

📍 Burlington, VT

- Built internal tool to help analyze and visualize user interaction with back-end products.

2015

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Data Science Intern
 Dealer.com

📍 Burlington, VT

- Worked with the product analytics team to help parse and visualize large stores of data to drive business decisions.

2015
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2014

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Data Artist In Residence
 Conduce

📍 Carpinteria, CA

- Envisioned, prototyped and implemented visualization framework in the course of one month.
 - Constructed training protocol for bringing third parties up to speed with new protocol.

2014

•

Software Engineering Intern
 Conduce

📍 Carpinteria, CA

- Incorporated d3.js to the company's main software platform.



TEACHING EXPERIENCE

- 2020

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Javascript for Shiny Users
 RStudio::conf 2020

- Served as TA for two day workshop on how to leverage Javascript in Shiny applications
 - Lectured on using R2D3 package to build interactive visualizations.⁶

2019

•

Data Visualization Best Practices
 DataCamp

- Designed from bottom up course to teach best practices for scientific visualizations.
 - Uses R and ggplot2.
 - In top 10% on platform by popularity.

I am passionate about education. I believe that no topic is too complex if the teacher is empathetic and willing to think about new methods of approaching task.

- 2019
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2018
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2017

Improving your visualization in Python
 DataCamp

 - Designed from bottom up course to teach advanced methods for enhancing visualization.
 - Uses python, matplotlib, and seaborn.
- 2018
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2017

Advanced Statistical Learning and Inference
 Vanderbilt Biostatistics Department

📍 Nashville, TN

 - TA and lectured
 - Topics covered from penalized regression to boosted trees and neural networks
 - Highest level course offered in department
- 2018

Advanced Statistical Computing
 Vanderbilt Biostatistics Department

📍 Nashville, TN

 - TA and lectured
 - Covered modern statistical computing algorithms
 - 4th year PhD level class
- 2017

Statistical Computing in R
 Vanderbilt Biostatistics Department

📍 Nashville, TN

 - TA and lectured
 - Covered introduction to R language for statistics applications
 - Graduate level class

SELECTED DATA SCIENCE WRITING

- 2019

Using AWK and R to Parse 25tb⁸
 LiveFreeOrDichotomize.com

 - Achieved 4,800x performance improvement for large-scale genomic data processing.
 - Reached top of HackerNews multiple times
- 2018

Classifying physical activity from smartphone data⁹
 RStudio Tensorflow Blog

 - Walk through of training a convolutional neural network to achieve state of the art recognition of activities from accelerometer data.
 - Contracted article.
- 2018

The United States of Seasons¹⁰
 LiveFreeOrDichotomize.com

 - GIS analysis of weather data to find the most 'seasonal' locations in United States
 - Used Bayesian regression methods for smoothing sparse geospatial data.

I regularly blog about data science and visualization on my blog LiveFreeOrDichotomize.com.⁷

- 2017 • **A year as told by fitbit¹¹**
LiveFreeOrDichotomize.com
 - Analyzing a full years worth of second-level heart rate data from wearable device.
 - Demonstrated visualization-based inference for large data.
- 2017 • **MCMC and the case of the spilled seeds¹²**
LiveFreeOrDichotomize.com
 - Full Bayesian MCMC sampler running in your browser.
 - Coded from scratch in vanilla Javascript.
- 2017 • **The Traveling Metallurgist¹³**
LiveFreeOrDichotomize.com
 - Pure javascript implementation of traveling salesman solution using simulated annealing.
 - Allows reader to customize the number and location of cities to attempt to trick the algorithm.



SELECTED PRESS (ABOUT)

- 2017 • **Great paper? Swipe right on the new ‘Tinder for preprints’ app¹⁴**
Science
 - Story of the app Papr¹⁵ made with Jeff Leek and Lucy D’Agostino McGowan.
- 2017 • **Swipe right for science: Papr app is ‘Tinder for preprints’¹⁶**
Nature News
 - Second press article for app Papr.
- 2016 • **The Deeper Story in the Data¹⁷**
University of Vermont Quarterly
 - Story on my path post graduation and the power of narrative.



SELECTED PRESS (BY)

- 2016 • **The Great Student Migration¹⁸**
The New York Times
 - Most shared NYT article of August 2016, demonstrating ability to create engaging UIs.
 - Used d3.js to realtime render 100 maps for personalized inspection for readers.

- 2016 • **Wildfires are Getting Worse, The New York Times¹⁹**
The New York Times
 - GIS analysis and modeling of fire patterns and trends
 - Data in collaboration with NASA and USGS
- 2016 • **Who's Speaking at the Democratic National Convention?²⁰**
The New York Times
 - Data scraped from CSPAN records to figure out who talked and past conventions.
- 2016 • **Who's Speaking at the Republican National Convention?²¹**
The New York Times
 - Used same data scraping techniques as Who's Speaking at the Democratic National Convention?
- 2016 • **A Trail of Terror in Nice, Block by Block²²**
The New York Times
 - Led research effort to put together story of 2016 terrorist attack in Nice, France in less than 12 hours.
 - Work won Silver medal at Malofiej 2017, and gold at Society of News and Design.



SELECTED PUBLICATIONS, POSTERS, AND TALKS

- 2020 • **Building a software package in tandem with machine learning methods research can result in both more rigorous code and more rigorous research**
ENAR 2020
 - Invited talk in Human Data Interaction section.
 - How and why building an R package can benefit methodological research
- 2020 • **Stochastic Block Modeling in R, Statistically rigorous clustering with rigorous code²³**
RStudio::conf 2020
 - Invited talk about new sbmR package²⁴.
 - Focus on how software development and methodological research can improve both benefit when done in tandem.
- 2020 • **PheWAS-ME: A web-app for interactive exploration of multimorbidity patterns in PheWAS²⁵**
Bioinformatics
 - Manuscript detailing application for the exploration of multimorbidity patterns in PheWAS analyses
 - See landing page²⁶ for more information.

- 2019 • **Charge Reductions Associated with Shortening Time to Recovery in Septic Shock**²⁷
Chest
• Authored with Wesley H. Self, MD MPH; Dandan Liu, PhD; Stephan Russ, MD, MPH; Michael J. Ward, MD, PhD, MBA; Nathan I. Shapiro, MD, MPH; Todd W. Rice, MD, MSc; Matthew W. Semler, MD, MSc.
- 2019 • **Multimorbidity Explorer | A shiny app for exploring EHR and biobank data**²⁸
RStudio::conf 2019
• Contributed Poster. Authored with Yaomin Xu.
- 2019 • **Taking a network view of EHR and Biobank data to find explainable multivariate patterns**²⁹
Vanderbilt Biostatistics Seminar Series
• University wide seminar series.
- 2019 • **Patient-specific risk factors independently influence survival in Myelodysplastic Syndromes in an unbiased review of EHR records**
Under-Review (copy available upon request.)
• Bayesian network analysis used to find novel subgroups of patients with Myelodysplastic Syndromes (MDS).
• Analysis done using method built for my dissertation.
- 2019 • **Patient specific comorbidities impact overall survival in myelofibrosis**
Under-Review (copy available upon request.)
• Bayesian network analysis used to find robust novel subgroups of patients with given genetic mutations.
• Analysis done using method built for my dissertation.
- 2018 • **R timelineViz: Visualizing the distribution of study events in longitudinal studies**
Under-Review (copy available upon request.)
• Authored with Alex Sunderman of the Vanderbilt Department of Epidemiology.
- 2017 • **Continuous Classification using Deep Neural Networks**³⁰
Vanderbilt Biostatistics Qualification Exam
• Review of methods for classifying continuous data streams using neural networks
• Successfully met qualifying examination standards
- 2015 • **Asymmetric Linkage Disequilibrium: Tools for Dissecting Multiallelic LD**
Journal of Human Immunology
• Authored with Richard Single, Vanja Paunic, Mark Albrecht, and Martin Maier.

- 2015 • **An Agent Based Model of Mysis Migration**³¹
International Association of Great Lakes Research Conference
• Authored with Brian O'Malley, Sture Hansson, and Jason Stockwell
- 2015 • **Declines of Mysis diluviana in the Great Lakes**
Journal of Great Lakes Research
• Authored with Peter Euclide and Jason Stockwell

LINKS

- 1: <https://www.nytimes.com/interactive/2016/08/26/us/college-student-migration.html>
- 2: https://livefreeordichotomize.com/2019/06/04/using_awk_and_r_to_parse_25tb/
- 3: <https://github.com/nstrayer/shinysense>
- 4: <http://nickstrayer.me/dataDayTexas/>
- 5: <https://ir.vanderbilt.edu/handle/1803/16394?show=full>
- 6: http://nickstrayer.me/js4shiny_r2d3/slides
- 7: <https://livefreeordichotomize.com/>
- 8: https://livefreeordichotomize.com/2019/06/04/using_awk_and_r_to_parse_25tb/
- 9: <https://blogs.rstudio.com/tensorflow/posts/2018-07-17-activity-detection/>
- 10: <https://livefreeordichotomize.com/2018/02/12/the-united-states-of-seasons/>
- 11: <https://livefreeordichotomize.com/2017/12/27/a-year-as-told-by-fitbit/>
- 12: <https://livefreeordichotomize.com/2017/10/14/mcmc-and-the-case-of-the-spilled-seeds/>
- 13: <https://livefreeordichotomize.com/2017/09/25/the-traveling-metallurgist/>
- 14: <https://www.sciencemag.org/news/2017/06/great-paper-swipe-right-new-tinder-preprints-app>
- 15: <https://jhubiostatistics.shinyapps.io/papr/>
- 16: <https://www.nature.com/news/swipe-right-for-science-papr-app-is-tinder-for-preprints-1.22163>
- 17: <https://www.uvm.edu/uvmnews/news/deeper-story-data>
- 18: <https://www.nytimes.com/interactive/2016/08/26/us/college-student-migration.html?smid=pl-share>
- 19: <https://www.nytimes.com/interactive/2016/07/25/us/wildfire-seasons-los-angeles.html>
- 20: <https://www.nytimes.com/2016/07/26/upshot/democrats-may-not-be-unified-but-their-convention-speakers-are.html>
- 21: <https://www.nytimes.com/2016/07/19/upshot/whos-not-speaking-how-this-republican-convention-differs.html?smid=pl-share>
- 22: <https://www.nytimes.com/interactive/2016/07/14/world/europe/trail-of-terror-france.html>
- 23: http://nickstrayer.me/rstudioconf_sbm
- 24: <https://tbilab.github.io/sbmR/>
- 25: <https://academic.oup.com/bioinformatics/advance-article-abstract/doi/10.1093/bioinformatics/btaa870/5922817?redirectedFrom=fulltext>
- 26: https://prod.tbilab.org/phewas_me_info/
- 27: <https://www.ncbi.nlm.nih.gov/pubmed/30419234>
- 28: http://nickstrayer.me/rstudioconf19_me-poster/
- 29: http://nickstrayer.me/biostat_seminar/
- 30: http://nickstrayer.me/qualifying_exam/

- 31: <https://www.semanticscholar.org/paper/An-Agent-Based-Model-of-the-Diel-Vertical-Migration-Strayer-Stockwell/40493c78e8ecf22bd882d17ec99fd913ec4b9820>